H M A Mohit *Chowdhury*

chowdhury's profile hmamohit@gmail.com h-m-a-mohit-chowdhury

PhD student in Computer Science and Engineering and Research Assistant with a focus on machine learning, computer vision—specifically for time-series forecasting, interpolation, diffusion, graph embedding, and clustering. I have experience in developing deep learning models, forecasting time series, creating data embeddings, clustering, and designing scalable systems. I am skilled in deploying solutions using Python, PyTorch, and cloud platforms, as well as building full-stack tools. I am passionate about improving AI research by combining strong algorithm design with practical applications in complex data areas.

Education

Ph.D. in Computer Science and Engineering
University of North Texas — *2025 - Present*
University of Colorado Colorado Springs (Transferred at UNT) — *2022 - 2025*
M.Sc. in Computer Science — **GPA: 3.867**
University of Colorado Colorado Springs — *2024*
B.Sc. in Computer Science and Engineering — **GPA: 3.71**
Ahsanullah University of Science and Technology — *2017*

Scientific Research (Google Scholar Profile)

Journal

- **Chowdhury, H. M. A. M.**, Fuller, M., & Oluwadare, O. (2025). *Robin: an advanced tool for comparative loop caller result analysis leveraging large language models*, NAR Genomics and Bioinformatics, Volume 8, Issue 1, March 2026, lqag009, https://doi.org/10.1093/nargab/lqag009
- **Chowdhury, H.M.A.M.**, Oluwadare, O. *EmbedTAD Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data.* Commun Biol 9, 7 (2026), https://doi.org/10.1038/s42003-025-09224-z
- Rohit, M., **Chowdhury, H. M. A. M.**, & Oluwadare, O. (2024). *ScHiCAtt: Hi-C Super Resolution Using Attention Methods for Single Cell*. Computational and Structural Biotechnology Journal, Volume 27, 978 - 991, https://doi.org/10.1016/j.csbj.2025.02.031
- Pinchuk, D., **Chowdhury, H. M. A. M.**, Pandeya, A., & Oluwadare, O. (2024). *HiCForecast: dynamic network optical flow estimation algorithm for spatiotemporal Hi-C data forecasting*. Bioinformatics, Volume 41, Issue 2, February 2025, https://doi.org/10.1093/bioinformatics/btaf030
- Houchens D, **Chowdhury H. M. A. M.**, Oluwadare O. (2024) *coiTAD: Detection of Topologically Associating Domains Based on Clustering of Circular Influence Features from Hi-C Data*. Genes, 15(10), 1293, https://doi.org/10.3390/genes15101293
- **Chowdhury, H. M. A. M.**, Boult, T., & Oluwadare, O. (2024). *Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness*. BMC Bioinformatics, 25(1), 123, https://doi.org/10.1186/s12859-024-05713-w
- Akpokiro, V., **Chowdhury, H. M. A. M.**, Olowofila, S., Nusrat, R., & Oluwadare, O. (2023). *CNNSplice: Robust models for splice site prediction using convolutional neural networks*. Computational and Structural Biotechnology Journal, Volume 21, 3210–3223, https://doi.org/10.1016/j.csbj.2023.05.031

Poster Presentation

- EmbedTAD: Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data. (Colorado AI Summit, 2025).
- Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness. (RECOMB 2024).
- Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness. (UCCS MLRD 2023).

Manuscript Reviewer

- Scientific Reports, Nature Portfolio
- BioData Mining, BMC
- IEEE International Conference on Bioinformatics and Biomedicine, IEEE

Skills

P. Languages:	Python, Java, C#, C, JavaScript	**Databases**:	MySQL, MSSQL
Frameworks:	PyTorch, cu (RAPIDS API), scikit-learn, Spring, .NET	**Deployment Tools**:	AWS, Docker, Terraform, git
		Operating Systems:	Linux, Windows, MacOS

Experience

Oluwadare Lab **2022 - Present**
Graduate Research Assistant

- Researching the development of novel algorithms in computer vision—specifically for time-series forecasting, interpolation, diffusion, graph embeddings, and clustering, to study 3D genome organization using Hi-C data, with a focus on chromatin loop prediction, TAD detection, and Hi-C data prediction.
- Developed a CNN-based deep learning model to predict splice sites in DNA sequences.
- Designed computer vision models to predict Hi-C contact maps for time-series analysis.
- Built computational models for TAD detection, including a graph-based framework with network embeddings and clustering.
- Conducted a study comparing chromatin loop detection methods to assess existing approaches.
- Built a web server for genomic analysis tools using Flask, HTML, CSS, JavaScript, and Docker to improve accessibility.
- Collaborated on projects, providing data support, research help, and assistance with server management.
- Reviewed manuscripts and co-authored publications in computational genomics and machine learning.

Otto International **2022 – 2022**
Solution Developer

- Designed and built AWS infrastructure for the Enterprise Service Bus (ESB) using Terraform. This setup allowed for automated and repeatable deployments.
- Fixed critical bugs and improved ESB performance by redesigning components in Java (Micronaut) and PostgreSQL.
- Developed and deployed CI/CD pipelines with Jenkins to improve release reliability.
- Worked closely with business analysts and engineering teams to adjust system requirements and successfully integrate their applications with the ESB.
- I helped with knowledge sharing and documentation to support long-term scalability and maintainability.

Dovetail Technology **2020 – 2022**
Software Engineer

- Designed and developed the Volmet system with C#, .NET Web API, and MSSQL.
- Deployed the entire system on the IIS Web Server for production use.
- Built Windows services to support the Volmet system, using a microservices-inspired architecture to improve reliability.
- Helped the frontend team by contributing to development and integration with AngularJS.
- Worked with the business analyst and automation team to simplify requirements and expand the development process.

IdeaScale Bangladesh **2019 – 2020**
Software Engineer

- Developed and integrated new features while updating existing ones to meet changing business needs using Java, Spring, MySQL, and Docker.
- Fixed bugs and improved platform stability to enhance user experience.
- Refactored old raw SQL queries into JPA and Querydsl, which simplified the code and made it easier to maintain.
- Worked with team members to ensure scalability and a smooth transition to new processes.

Netweaver Software Limited **2017 – 2019**
Software Engineer

- Designed and developed two e-commerce platforms using Java, Spring Boot, MySQL, SAPUI5, and JavaScript.
- Built a comprehensive directory for doctor and hospital information, using Node.js, Vue.js, and PostgreSQL.
- Focused on responsive design, scalability, and seamless integration between frontend and backend components.
- Worked with different teams to clarify requirements and deliver ready-to-use applications.

Awards

- CENG Anticipated Award, 2025 - University of North Texas.
- Graduate School Travel Award, 2024 - University of Colorado Colorado Springs.
- Tuition Matching Grant Award, 2023-2024 - University of Colorado Colorado Springs.
- Tuition Matching Grant Award, 2022-2023 - University of Colorado Colorado Springs.

Certificates

- Tableau Fundamentals.
- Responsible Conduct of Research (RCR-mini).

Activities

- Mentor, UR2PHD at UNT.
- Participant, Open House 2025 at UCCS.
- President, Bangladeshi Student Association (BSA) at UCCS (2024-2025).
- Participant, Cool Science Festival 2024 at UCCS.
- Session Chair, NSF REU Summer Research 2023 at UCCS.
- Technical Assistant, NSF REU Summer Research 2023 at UCCS.
- Participant, Cool Science Festival 2023 at UCCS.
- Conducted a basic Linux and Docker hands-on class in NSF REU Summer Research 2023 at UCCS.
- Participant, Cool Science Festival 2022 at UCCS.
- Organized *Cumilla Victoria Govt. College Reunion 2016*.